IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

  In this project, I followed a standard data science workflow. The process started with data collection and preparation (ETL), followed by exploratory data analysis (EDA) to identify key patterns. Finally, several machine learning models (Logistic Regression, SVM, Decision Tree, K-Nearest Neighbors) were built and evaluated to find the best-performing model for predicting Falcon 9 first-stage landing outcomes.

- Summary of all results

  After conducting data collection, processing, and analysis, we developed a predictive model with 83% accuracy for determining whether a Falcon 9 booster would successfully land. These insights can help stakeholders estimate launch costs and improve reusability strategies.

Introduction

The commercial space industry is evolving rapidly, with companies like Virgin Galactic, Rocket Lab, and Blue Origin competing in the market. SpaceX has led the innovation by making rocket reusability a key feature, drastically reducing costs (from competitors' $165M per launch to SpaceX's $62M). Understanding and predicting the reusability of the Falcon 9 first-stage is crucial for optimizing costs and operations.

This is done by reusing the first stage thus reducing costs as it’s the most expensive part Predicting landing helps estimate launch cost. The objective is to determine launch costs and predict first stage landing success.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Using the SpaceX REST API and web scraping, collected, filtered, and cleaned launch data to predict Falcon 9 first-stage landing outcomes.
- Perform data wrangling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification mode

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

Sources Used:

• SpaceX REST API: Extracted data including mission launch dates, rocket, types, payloads, and landing outcomes.

Data Retrieval Techniques:

- • API Calls: Implemented GET requests to fetch real-time mission data.
- • Data Cleaning: Addressed missing values, standardized formats, and handled anomalies.
- • Data Storage: Stored data locally in CSV files for efficient manipulation and analysis.

Place your flowchart of SpaceX API calls here

Data Collection - Scraping

Web Scraping Process:

- Implemented a systematic approach to webscraping using key phrases and logical structures.
- Ensured data quality by cleaning and verifying extracted information before analysis.

Place your flowchart of web scraping here

Data Wrangling

- **Describe how data were processed**

[Load Raw Data] → [Remove Duplicates] → [Format Columns] → [Handle Missing Values] →[Filter Falcon 9 Launches] → [Final Clean Dataset]

EDA with Data Visualization

What charts were plotted and why I used those charts ?

- Scatter Plots : Flight Number vs. Launch Site, Payload vs. Launch Outcome.
- Bar Charts : Success rate by orbit type.
- Time Series Analysis : Yearly success rate trends.

EDA with SQL

- Calculating total or average payload masses carried by rockets.
- Ranking landing outcomes based on mission success or other criteria.
- Organizing data to uncover patterns or trends in space exploration.
- Identifying the launch cities for various space missions.

Build an Interactive Map with Folium

Map Objects Added:

- Included markers to pinpoint specific locations of interest.
- Added circle markers to highlight regions with visual emphasis on areas of
- significance.
- Used lines to connect related points, such as routes or paths between markers.

Build a Dashboard with Plotly Dash

Components of Dashboard

- Scatter Plots: Representing relationships between variables, with hover-to-inspect details.
- Bar Charts: Summarizing categorical data for comparative analysis.
- Interactive Filters: Allowing users to adjust visualizations based on specific criteria (e.g., date range, category).

Predictive Analysis (Classification)

Model Development Process:

- Building the Model:
  - Selected suitable algorithms for classification tasks (e.g., logistic regression, decision trees, KNN, etc.).
  - Preprocessed the dataset to ensure quality and consistency.
  - Used pipeline creation for streamlined model implementation.
- Evaluating the Model:
  - Applied key metrics like accuracy, precision, recall, F1 score, and ROC curves.
  - Compared performance across different algorithms to identify strengths and weaknesses.
- Improving the Model:
  - Tuned hyperparameters to optimize performance (e.g., max depth in decision trees, regularization in SVMs).
  - Leveraged techniques like cross-validation for robust evaluation.

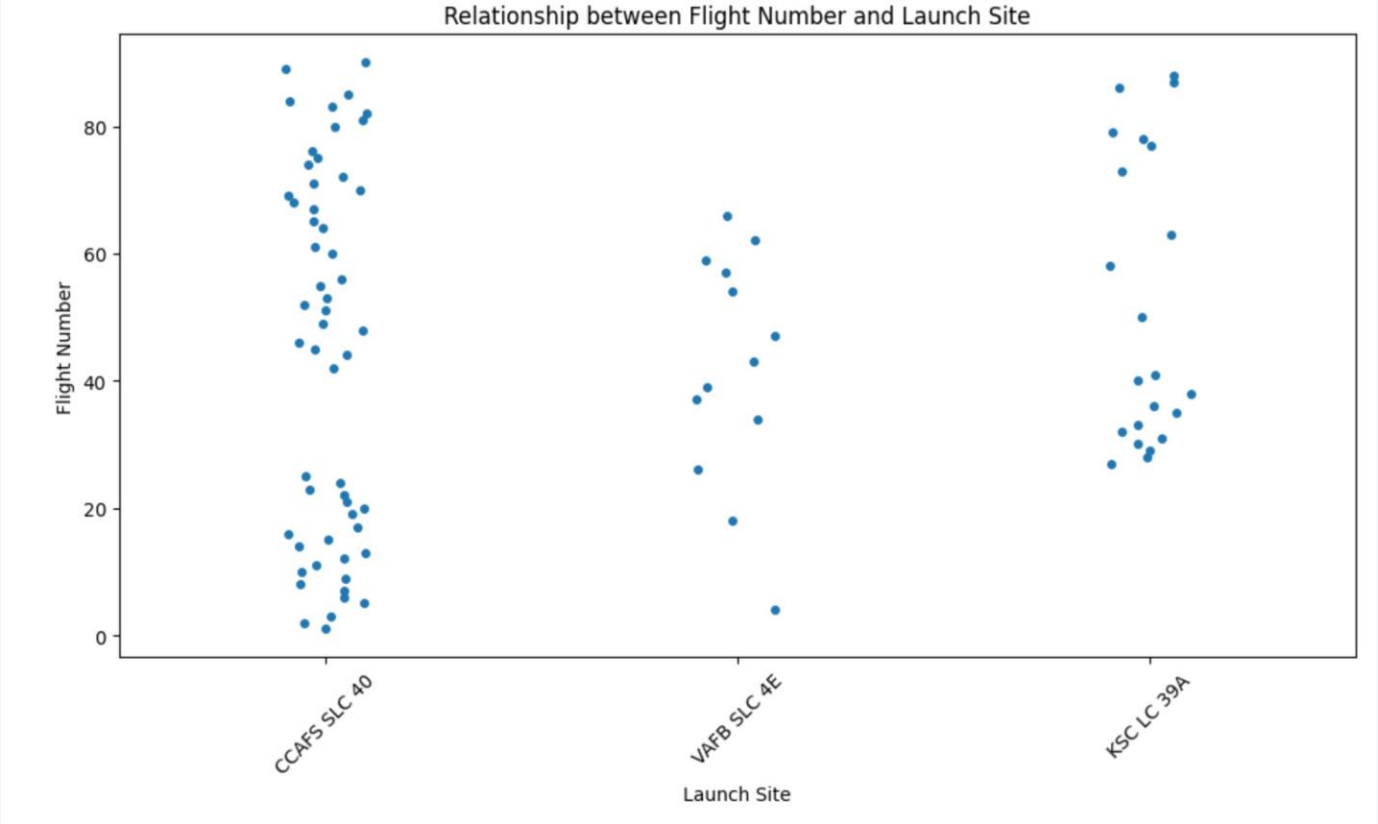
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

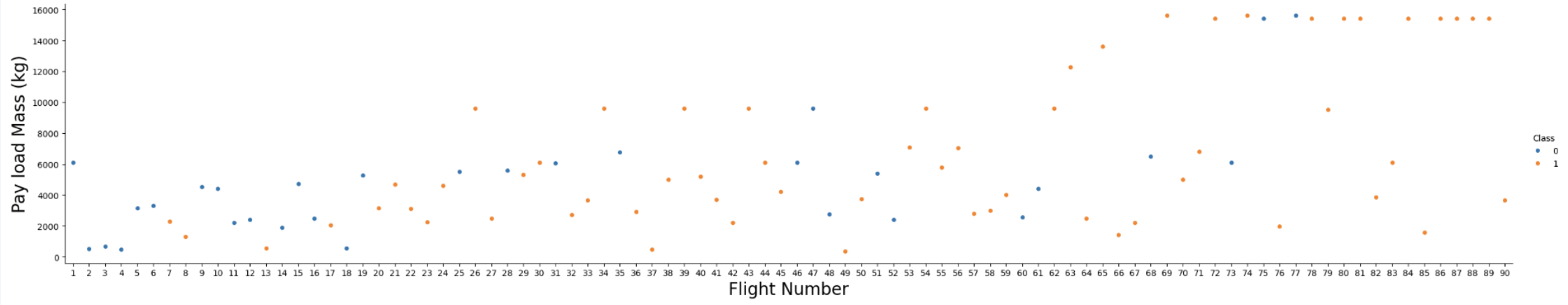
Insights drawn from EDA

Flight Number vs. Launch Site

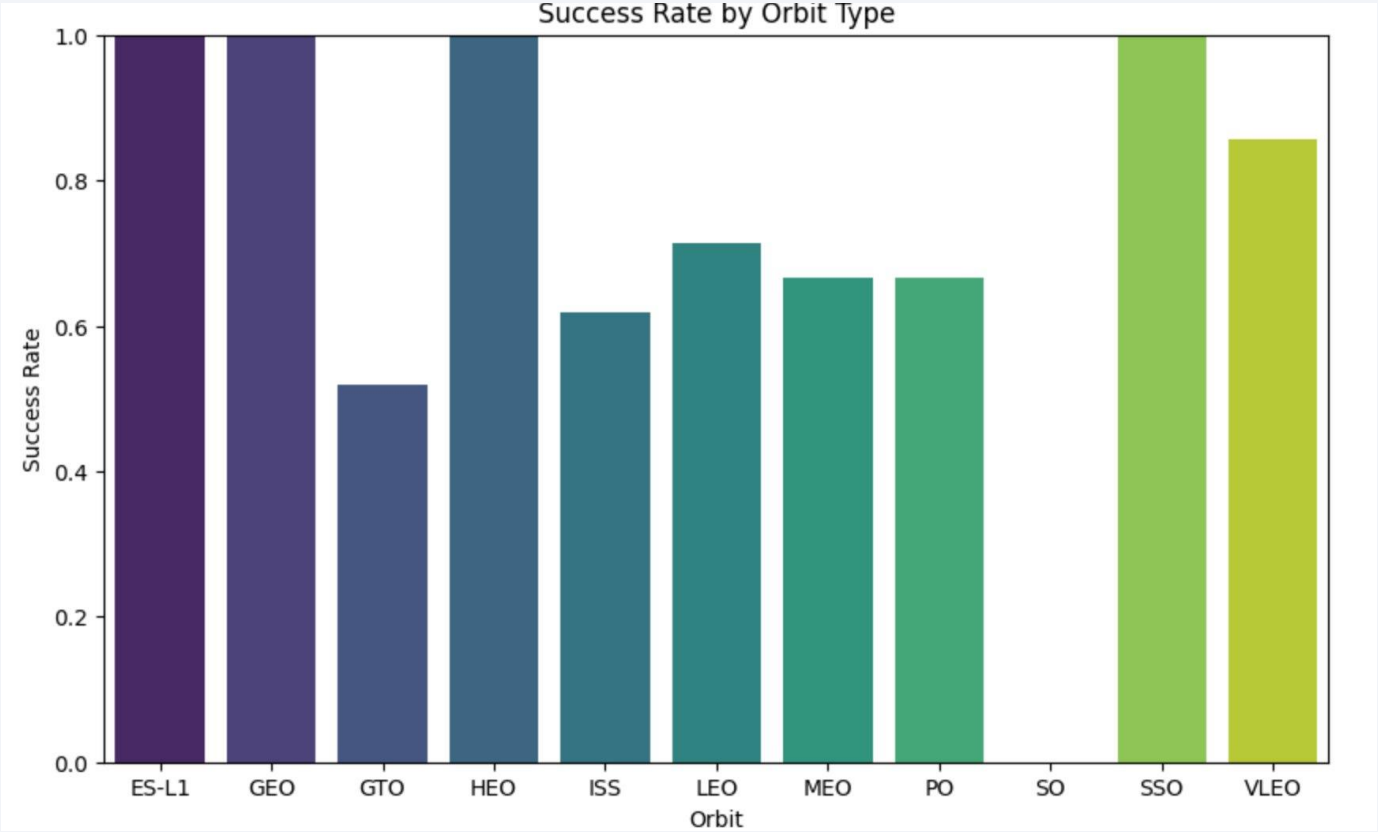


- The most launch attempts was in the Launch site: CCAFS SLC 40, followed by KSC LC 39A and ended byVAFB SLC 4E

Payload vs. Launch Site

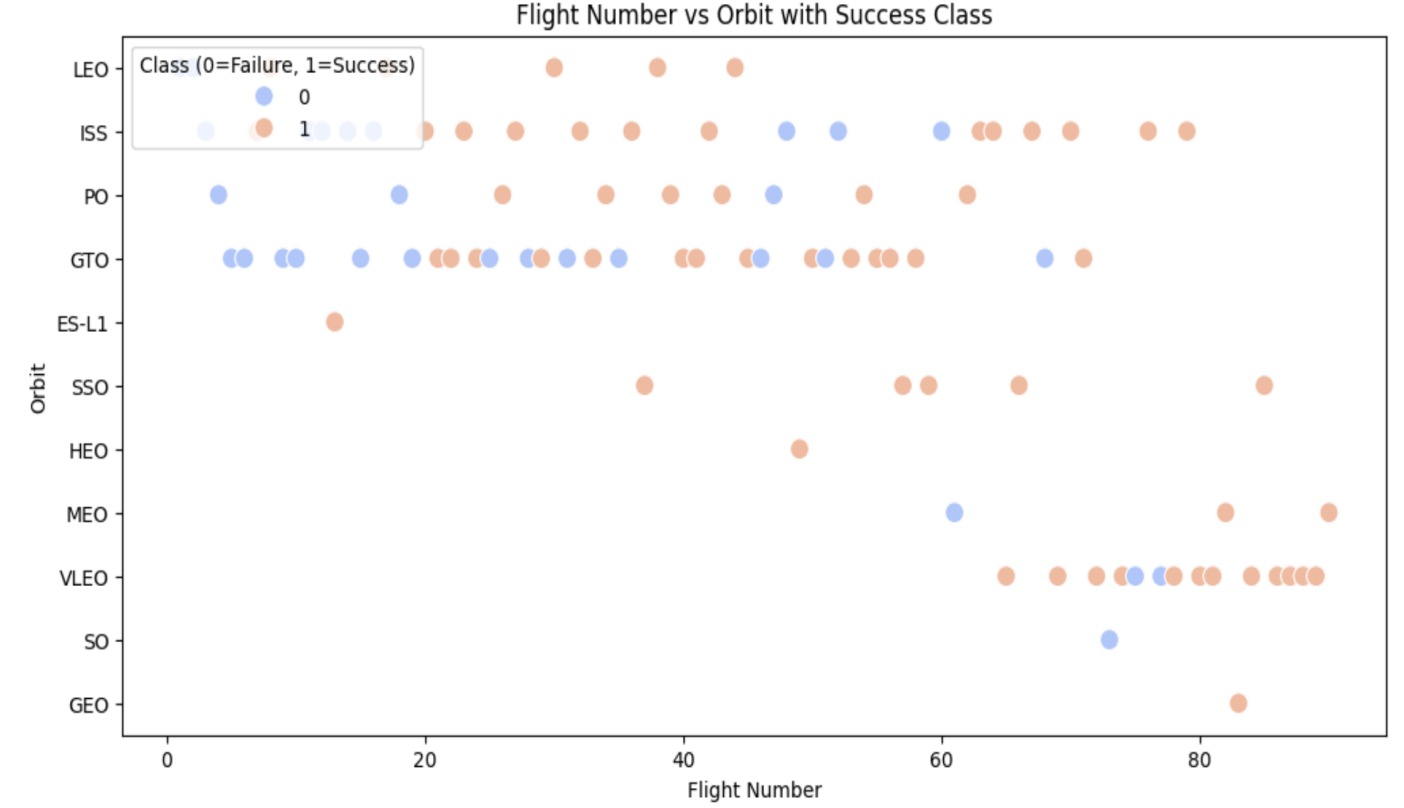


Success Rate vs. Orbit Type

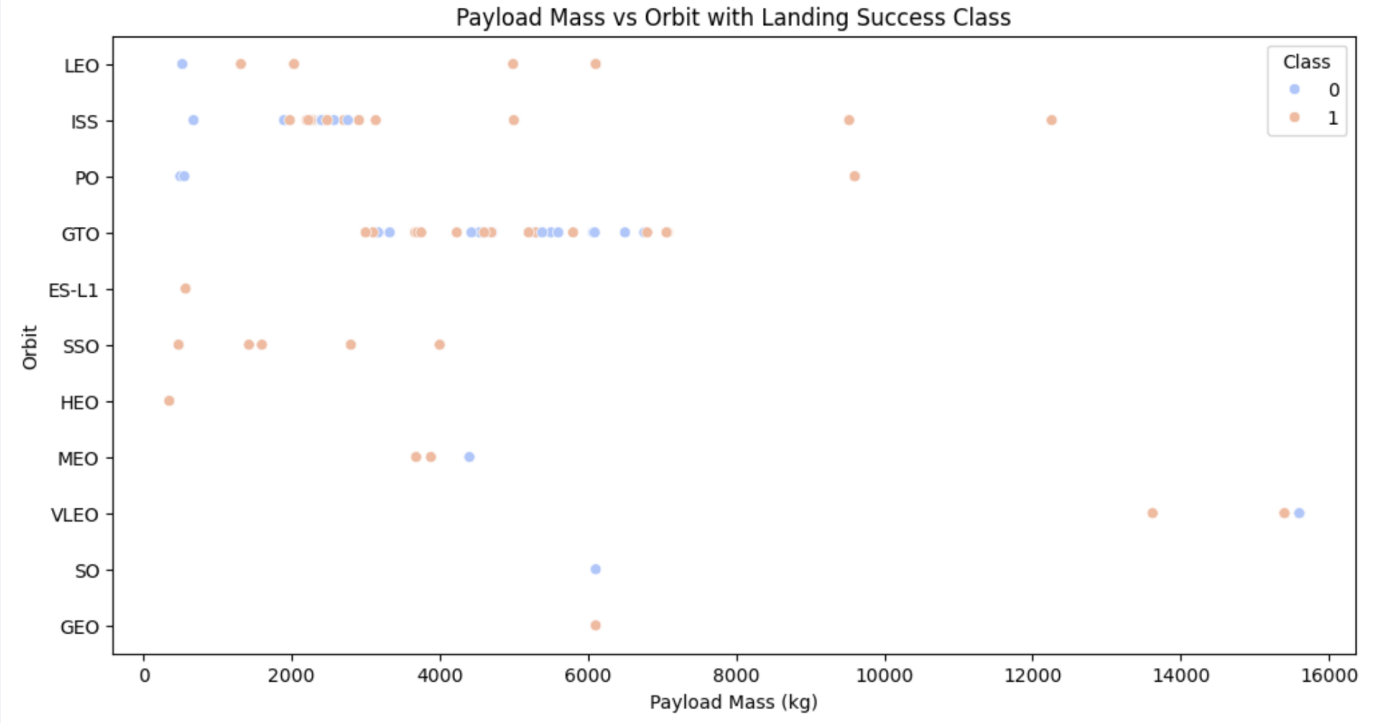


- The orbits which have the highest success rate: ES-L1, GEO, HEO, SSO

Flight Number vs. Orbit Type



Payload vs. Orbit Type



With heavy payloads the successful landing rate are more for GTO, LEO and SSO.

Launch Success Yearly Trend

The success rate started increasing from 2013 until 2020

All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

The names of the unique launch sites

Launch Site Names Begin with 'CCA'

ate	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
10-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
10-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
12-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
12-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
13-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

This query give the name of launch site that begin by ‘CCA’

Total Payload Mass

total_payload_by_NASA_CRS
48213

The total payload mass carried by boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

avg_payload_F9_v1_1
2928.4

The average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

first_success_ground_pad
2015-12-22

The date of the first successful landing outcome on ground pad is December 22, 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

List of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

The total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

List of names of the booster which have carried the maximum payload mass

2015 Launch Records

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

List of failed landing_outcomes in drone ship, their booster versions, and launch sitenames for in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

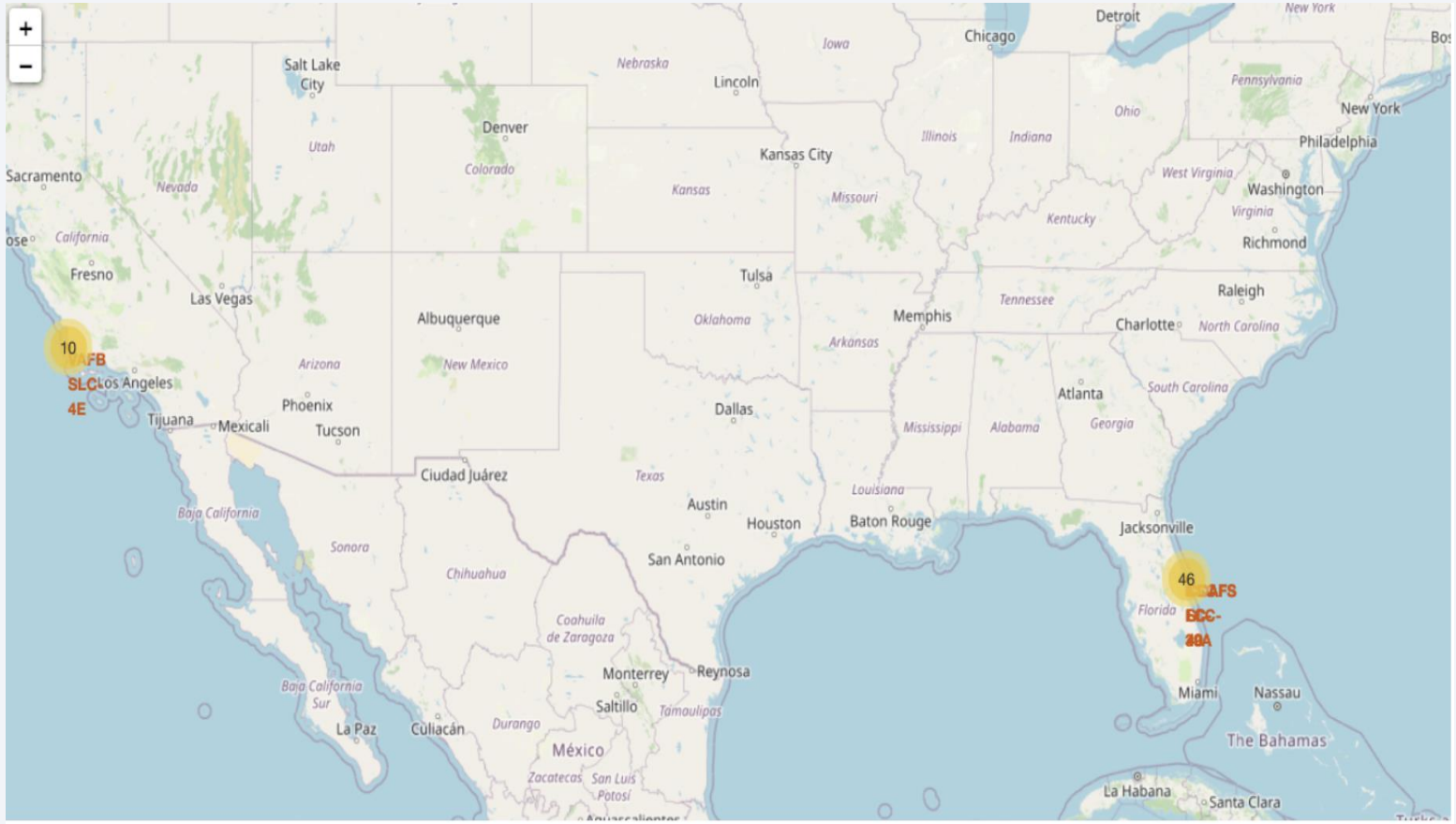
Landing_Outcome	count_outcome
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

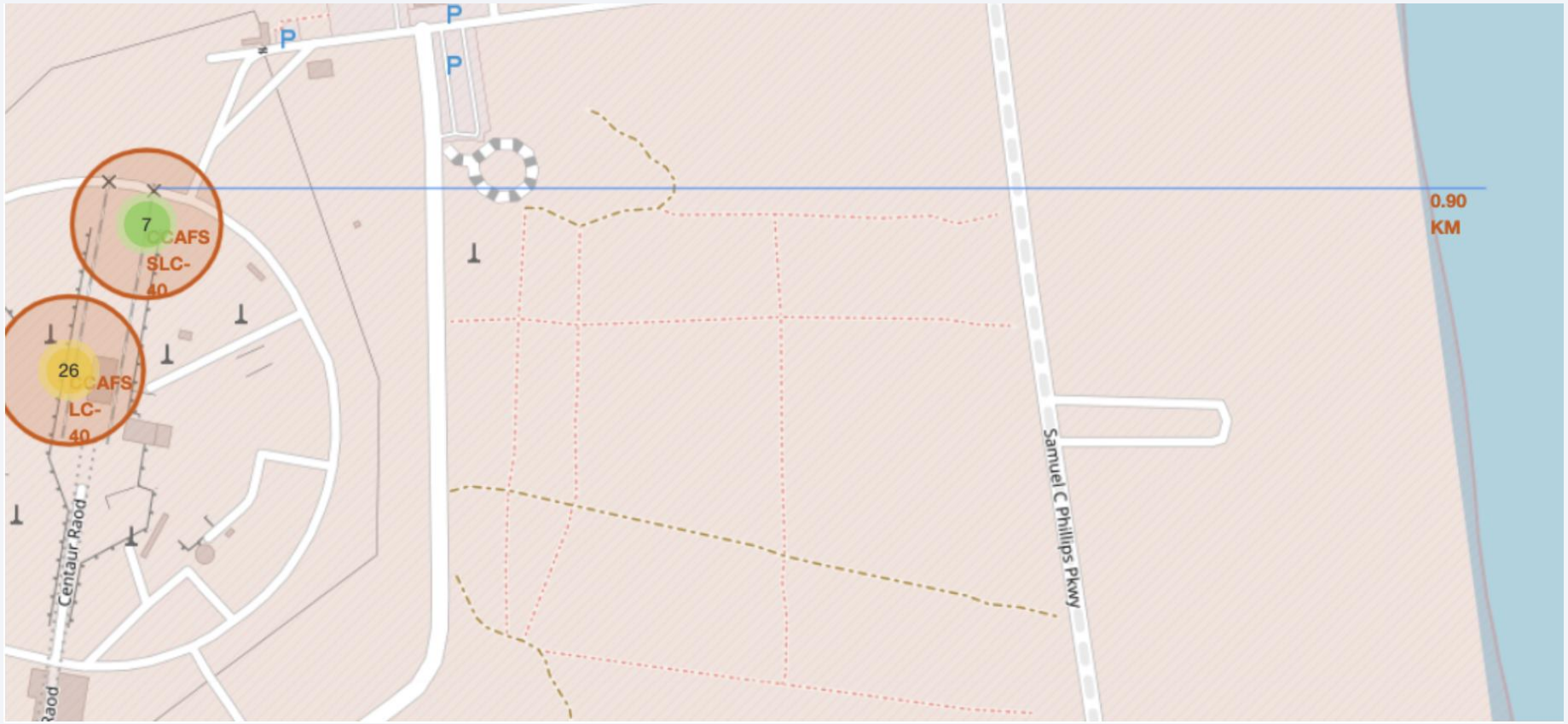
Section 3

Launch Sites Proximities Analysis

Map with marked launch sites



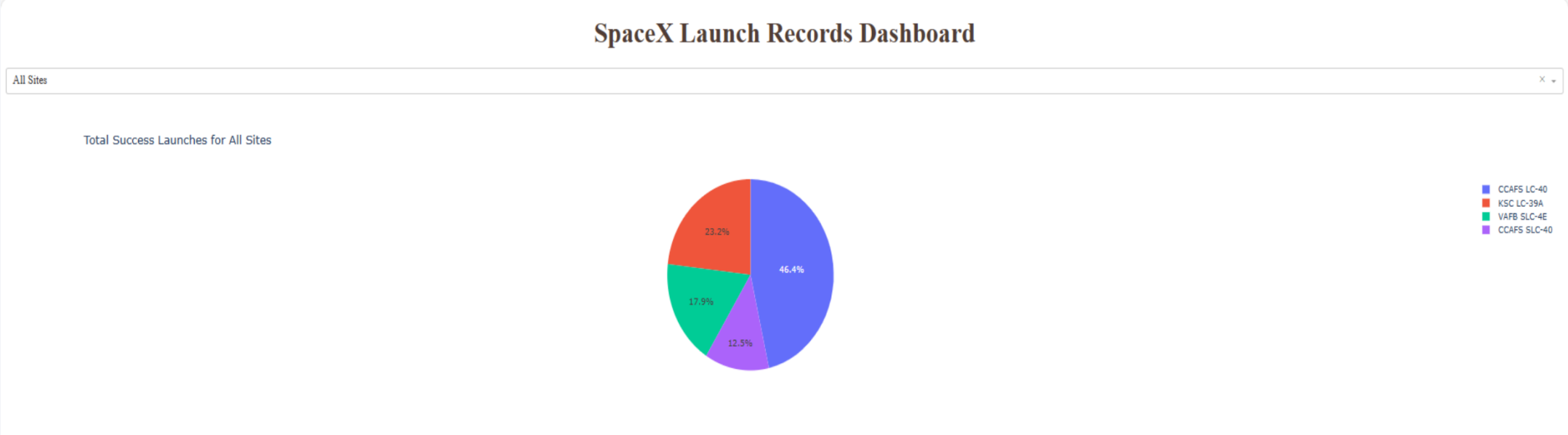
Proximities of launch site



Section 4

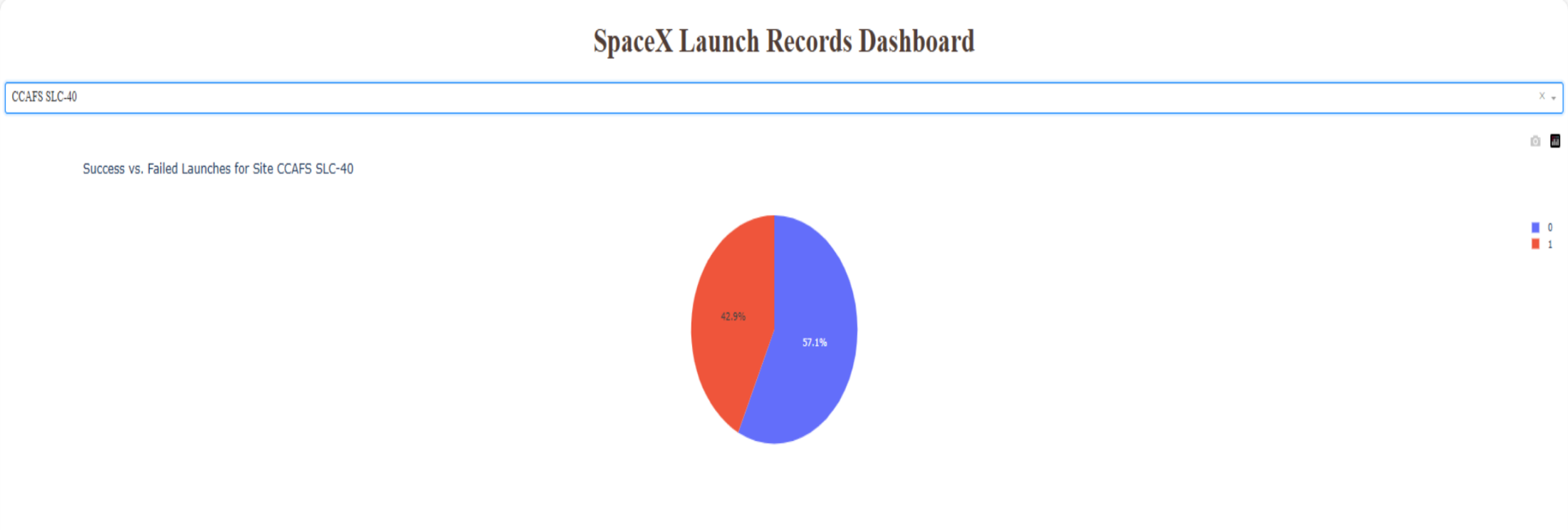
Build a Dashboard with Plotly Dash

Total Success Launch for all sites

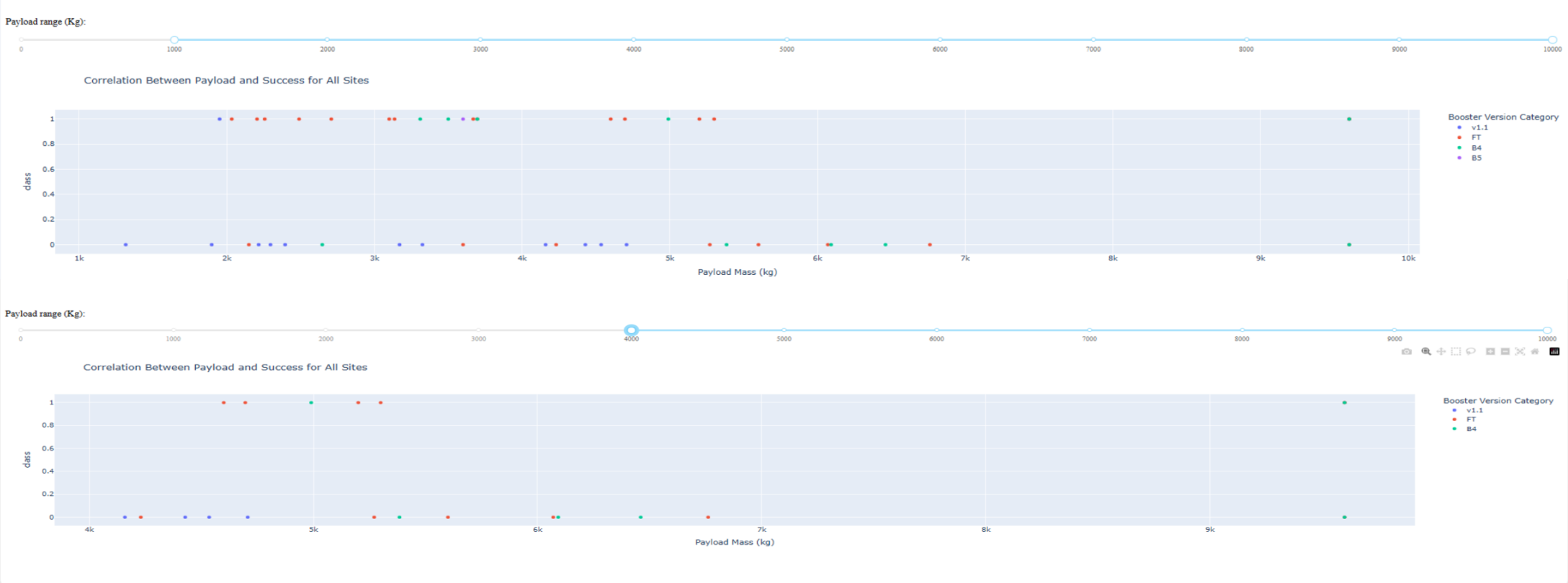


- CCAFS site has the highest launches

Highest launches on a site



Payload vs Launch Outcome



The booster version V1 -1 have the highest success rate

Section 5

Predictive Analysis (Classification)

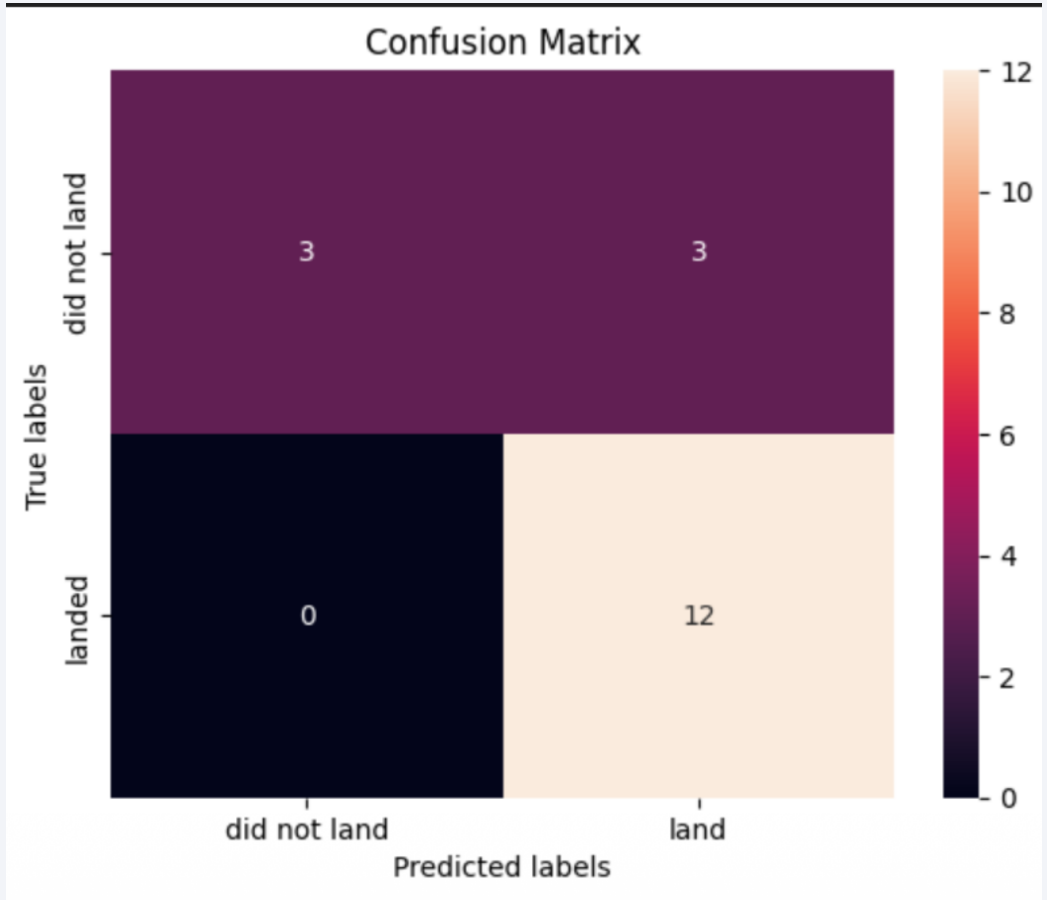
Classification Accuracy

Accuracy percentages for each model:

- Logistic Regression: 84.64%
- Decision Tree: 88.92%
- SVM: 84.82 %
- K Nearest Neighbors: 84.82%

- Conclusion: Decision Tree Classifier achieved the highest classification accuracy at 88.92%, making it the best-performing model.

Confusion Matrix



It performed 12 True Positive, 3 False Negative, 0 False Positive and 3 True Negative. Therefore, it only made mistakes by predicted three success land instead of predict fail.

Conclusions

- Developed advanced models, achieving up to 88% accuracy (Decision Tree).
- Designed interactive dashboards showcasing data insights effectively.
- Created a detailed Folium map for spatial analysis.

Thank you!